Copula Entropy

Theory and Applications

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Lecture 2: Theory of Copula Entropy (2)

1. Physical Meaning
2. Estimation
3. Growth of Theory

Physical Meaning of Copula Entropy

- **Laws of Thermodynamics**
  - First law: Conservation of energy
    Energy cannot be created or destroyed and only changes from one form to another.
    $$\Delta U = Q - W \tag{1}$$
  - Second law: Entropy increase
    The total disorder of an isolated system always goes up over time.
    $$\delta Q = TdS \tag{2}$$
  - Third law: Absolute zero limit
    As temperature approaches absolute zero (0 K), a system's entropy approaches a constant minimum.

Physical Meaning of Copula Entropy

- **Physical meaning of copula entropy**
  entropy reduction due to increasing order of thermodynamic systems

$$H_c(\mathbf{x}) \leq 0 \tag{3}$$

Physical Meaning of Copula Entropy

- **Systems that keep order**
  - Earth
    constantly receiving energy from Sun to keep earth system evolving
  - Society
    receiving energy from natural environment to keep society running
  - Life
    getting food and energy from outside to keep alive

Copula Entropy: Estimation

- **Parametric** Estimation by Definition

$$H_c(\mathbf{x}) = -E(\log c(\mathbf{u}; \alpha)). \tag{4}$$

1. estimating empirical copula density $\mathbf{u}$ based on rank statistic

$$u_i = F_i(x_i) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}(x_t^i \leq x_i) \tag{5}$$

2. estimating the parameter α of copula density function $c(\cdot)$ with maximum likelihood method
3. calculating according to (4)

Copula Entropy: Estimation

- **Non-Parametric** Estimation Method[1]
  1. estimating empirical copula density $\mathbf{u}$ based on rank statistic

$$u_i = F_i(x_i) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}(x_t^i \leq x_i) \tag{6}$$

  2. estimating copula entropy with the kNN entropy estimation method
- Advantages
  - distribution-free, non-parametric
  - tuning-free, insensitive to parameters
  - good convergence
  - easy to implement
  - low computation burden

[1] Jian Ma and Zengqi Sun. "Mutual Information Is Copula Entropy". In: *Tsinghua Science & Technology* 16.1 (2011). See also arXiv preprint arXiv:0808.0845, 2008, pp. 51–54.

Copula Entropy: Estimation

- **Conditional Mutual Information** Estimation

can be done with copula entropy estimators according to (7).

$$I(x; y|z) = H_c(x, z) + H_c(y, z) - H_c(x, y, z). \quad (7)$$

1. estimating the CE terms in (7) with CE estimator
2. calculating the result

Growth of Theory

Classical Information Theory

- CE based MI estimation, such as GCMI[2], MICE[3], Vector Copula based MI estimation[4], Vine copula based MI and CMI estimation[5], semi-parametric MI estimation[6], kernel based CE estimation[7], non-parametric CE estimation[8];

[2] Robin A.A. Ince et al. "A statistical framework for neuroimaging data analysis based on mutual information estimated via a gaussian copula". In: *Human Brain Mapping* 38.3 (2017), pp. 1541–1573.

[3] Adrien Cortes. "Accélération des techniques d' analyse de facteurs par méthodes d' apprentissage statistique et application à un GSE". Mémoire. Institut des Sciences Financiéres et d' Assurance, 2023.

[4] Yanzhi Chen et al. "Neural Mutual Information Estimation with Vector Copulas". In: *Advances in Neural Information Processing Systems*. 2025.

[5] 王洪安. "上肢运动的肌间耦合分析方法研究". 硕士学位论文. 杭州电子科技大学, 2021, Houman Safaai. "Amortized Vine Copulas for High-Dimensional Density and Information Estimation". In: *Advances in Neural Information Processing Systems* (2026). See also arXiv:2604.20568. arXiv: 2604.20568 [cs.LG]. URL: https://arxiv.org/abs/2604.20568.

[6] M. Mohammadi, M. Hashempour, and M. Emadi. "Semiparametric estimation of mutual information for elliptical copulas". In: *The 7th Seminar on Copula Theory and its Applications*. 2023, p. 44.

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Growth of Theory

Classical Information Theory

- asymmetric MI estimation[9], CE^2[10], Bayesian dependence measure[11], and tail dependence measure[12];

[9] Soumik Purkayastha and Peter X.K. Song. "Asymmetric predictability in causal discovery: an information theoretic approach". In: *arXiv preprint arXiv:2210.14455* (2022).

[10] Lin Liu, Cong Hu, and Xiao-Jun Wu. "CE^2: A Copula Entropic Mutual Information Estimator for Enhancing Adversarial Robustness". In: *Pattern Recognition and Computer Vision*. Springer Singapore, 2024, pp. 163–174.

[11] Guillaume Marrelec and Alain Giron. "An Inferential Measure of Dependence Between Two Systems Using Bayesian Model Comparison". In: *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 55.3 (2025), pp. 1671–1683. DOI: 10.1109/TSMC.2024.3509025.

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Growth of Theory

Classical Information Theory

- CE based directed information decomposition[13];

$$DI(X^n \to Y^n) = \sum_{i=1}^{n} [I(X^i, Y^i) - I(X^i, Y^{i-1})] - I(Y^n) \tag{8}$$

[13] Aleksander Wieczorek and Volker Roth. "Causal Compression". In: *arXiv preprint arXiv:1611.00261* (2016),
Dinu Johannes Kaufmann. "Semi-parametric Gaussian Copula Models for Machine Learning". PhD thesis. University of Basel, 2017.

Growth of Theory

Classical Information Theory

- CE based maximal entropy[14] and maximal Tsallis CE[15].

[14] 進藤晋. "Copula Entropy の最大化". In: 日本 *OR* 学会 *2011* 年度秋季研究Ⓕ表会. 2011, pp. 362–363, Milan Bubák and Mirko Navara. "Fitting Copulas with Maximal Entropy". In: *Entropy* 27.1 (2025), p. 87. ISSN: 1099-4300. DOI: 10.3390/e27010087. URL: https://www.mdpi.com/1099-4300/27/1/87, Seyedeh Azadeh Fallah Mortezanejad et al. "Non-Parametric Multivariate Control Chart Using Copula Entropy". In: *Sankhya B* 87 (2025), pp. 468–518. URL: https://doi.org/10.1007/s13571-025-00374-y.

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Growth of Theory

Graph Theory

- CE based graph matching[16].

[16] Francisco Escolano et al. "The mutual information between graphs". In: *Pattern Recognition Letters* 87 (2017), pp. 12–19. DOI: https://doi.org/10.1016/j.patrec.2016.07.012. URL: https://www.sciencedirect.com/science/article/pii/S016786551630174X, Manuel Curado Navarro. "Structural Similarity: Applications to Object Recognition and Clustering". PhD thesis. Universitat d'Alacant, 2018.

Growth of Theory

Information Bottleneck

- CE based information bottleneck computation[17].

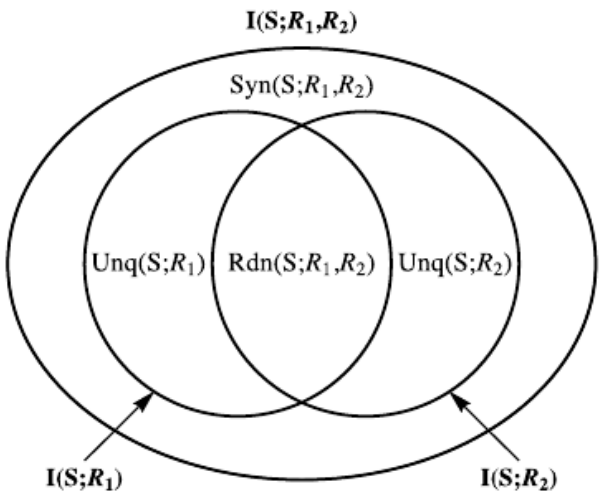
$$\mathcal{L}(\tilde{x}; x, y) = I(\tilde{X}; X) - \beta I(\tilde{X}; Y) \tag{9}$$

[17] Mélanie Rey and Volker Roth. "Meta-Gaussian information bottleneck". In: *Advances in Neural Information Processing Systems* 25 (2012), Mélanie Rey. "Copula Models in Machine Learning". PhD thesis. University of Basel, 2015, Mario Wieser et al. "Inverse Learning of Symmetries". In: *Advances in Neural Information Processing Systems*. Vol. 33. 2020, pp. 18004–18015, Mario Wieser et al. "Learning Sparse Latent Representations with the Deep Copula Information Bottleneck". In: *International Conference on Learning Representations*. 2018. URL: https://openreview.net/forum?id=Hk0wHx-RW, Mario Wieser. "Learning Invariant Representations for Deep Latent Variable Models". PhD thesis. University of Basel, 2020.

Growth of Theory

Partial Information Decomposition

- unique information estimation[18], synergy estimation[19], and Ω information estimation[20].



[18] Ari Pakman et al. "Estimating the Unique Information of Continuous Variables in Recurrent Networks". In: *Advances in Neural Information Processing Systems* 34 (2021), pp. 20295–20307.

[19] Vlad-Bogdan Coroian, Pedro Mediano, and Tolga Birdal. *Scaling up synergy estimators*. Tech. rep. Imperial College London, 2024.

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Growth of Theory

Information Geometry

- copula variational Bayes algorithm[21].

$$D_{KL}(\tilde{p}(\mathbf{x})|p(\mathbf{x})) = D_{KL}(\tilde{c}(\mathbf{u})|c(\mathbf{u})) + \sum_i D_{KL}(\tilde{p}_i(x_i)|p_i(x_i)) \tag{10}$$

$$\geq \sum_i D_{KL}(\tilde{p}_i(x_i)|p_i(x_i)) \tag{11}$$

[21] Robert Zimmerman. *Copulas and Information Geometry: Strange Bedfellows?* Tech. rep. URL:https://rob-zimmerman.ca/files/projects/STA4531_Copulas_IG_Strange_Bedfellows.pdf. University of Toronto, 2025.

Growth of Theory

Optimal Transport

- entropic optimal transport[22].

$$D_{EOT}(\epsilon) = \min_{P(X,Y)} E[d(X,Y)] + \epsilon I(X,Y) \tag{12}$$

[22] Giuseppe Serra. "Goal-oriented Semantic Communication : A Rate Distortion Perception Approach". PhD thesis. Sorbonne Université, 2026, Giuseppe Serra, Photios A Stavrou, and Marios Kountouris. "Copula-based estimation of continuous sources for a class of constrained rate-distortion functions". In: *2024 IEEE International Symposium on Information Theory (ISIT)*. IEEE. 2024, pp. 1089–1094.

Growth of Theory

Category Theory

- copula operad[23].

[23] Xuexing Lu. "Copula Operad and Copula Entropy". In: *arXiv preprint arXiv:2609.20512* (2026).

Growth of Theory

Copula Theory

- copula function estimation[24], and circular-linear copulas[25];

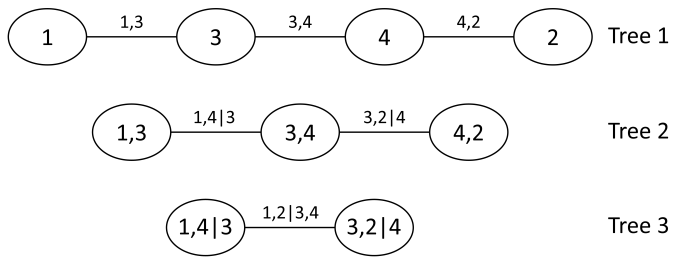
[24] Longxia Qian et al. "A New Estimation Method for Copula Parameters for Multivariate Hydrological Frequency Analysis With Small Sample Sizes". In: *Water Resources Management* 36.4 (2022), pp. 1141–1157. ISSN: 1573-1650. URL: https://doi.org/10.1007/s11269-021-03016-w, Rafael Calsaverini and Renato Vicente. "An information-theoretic approach to statistical dependence: Copula information". In: *EPL (Europhysics Letters)* 88.6 (2009), p. 68003. DOI: 10.1209/0295-5075/88/68003. URL: https://doi.org/10.1209/0295-5075/88/68003, Rafael S. Calsaverini. "Tópicos em Mecânica Estatística de Sistemas Complexos". PhD thesis. Universidade de São Paulo, 2013.

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Growth of Theory

Copula Theory

- Vine copula model selection[26].



[26] Fadhah Amer Alanazi. "Truncating Regular Vine Copula Based on Mutual Information: An Efficient Parsimonious Model for High-Dimensional Data". In: *Mathematical Problems in Engineering* 2021 (2021), p. 4347957. URL: https://doi.org/10.1155/2021/4347957, 王念鸽. "基于互信息的 Vine Copula 模型的高频数据投资组合风险测度研究". 硕士学位论文. 浙江财经大学, 2023, Jin Liu et al. "Vine-Copula Seismic Functionality Evaluation Method of Medical Room Systems Based on Shaking Table Tests". In: *Earthquake Engineering & Structural Dynamics* 54.6 (2025), pp. 1499–1519, Edith Kovács and Tamás Szántai. "Vine copulas as a mean for the construction of high dimensional probability distribution associated to a Markov Network". In: *4th Workshop on Vine Copula Distributions and Applications*. See also arXiv prerpint arXiv:1105.1697. 2011, Dániel Pfeifer. "Matrix-based vine copula building algorithms". MA thesis. Budapest University of Technology and Economics, 2022, Houman Safaai and Alessandro Marin Vargas. "Dynamic Vine Copulas: Detecting and Quantifying Time-Varying Higher-Order Interactions". In: *arXiv prerprint arXiv:2605.03061* (2026), Tobias Wolf, Andreas Wirsen, and Tobias Damm. "Information-Theoretic Regularization for High-Dimensional Vines: Mitigating Overfitting and Accelerating Estimation in Small Sample Regimes". In: *2026 29th International Conference on Information Fusion (FUSION)*. 2026, pp. 1–8. DOI: 10.23919/FUSION67272.2026.11726503.

Growth of Theory

Statistics

- ordinal majorization entropy[27].

[27] Kevin Fathi. "Ordinal Majorization Entropy (r-OME): Regime- Conditional Ordinal Aggregation with Dynamic Inequality Equilibrium, a Non-Entropic Stabilized- Regime Module, and Diffusion Diagnostics". Version v3. In: *Zenodo* (Jan. 2025). DOI: 10.5281/zenodo.18292593. URL: https://doi.org/10.5281/zenodo.18292593.

Growth of Theory

Causal Analysis

- causal compression[28], causal structure learning[29], LiNGAM[30], and time series causal network construction[31].

[28] Aleksander Wieczorek and Volker Roth. "Causal Compression". In: *arXiv preprint arXiv:1611.00261* (2016).

[29] Leonie Bossemeyer. "Machine Learning for Causal Discovery with Applications in Economics". MA thesis. Ludwig-Maximilians-Universität München, 2021, Marvin Lasserre, Régis Lebrun, and Pierre-Henri Wuillemin. "Learning Continuous High-Dimensional Models using Mutual Information and Copula Bayesian Networks". In: *Thirty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2021*. AAAI Press, 2021, pp. 12139–12146. URL: https://ojs.aaai.org/index.php/AAAI/article/view/17441, Marvin Lasserre. "Apprentissages dans les réseaux bayésiens à base de copules non-paramétriques". PhD thesis. Sorbonne Université, 2022. URL: https://tel.archives-ouvertes.fr/tel-03647090, Wenyu Wang and Yaping Wan. "Causal Discovery for Cross-Sectional Data Based on Super-Structure and Divide-and-Conquer". In: *2026 International Joint Conference on Neural Networks (IJCNN)*. 2026, pp. 1–7. DOI: 10.1109/IJCNN69854.2026.11730080.

[30] Joe Suzuki and Tian-Le Yang. "Generalization of LiNGAM that allows confounding". In: *2024 IEEE International Symposium on Information Theory (ISIT)*. IEEE. 2024, pp. 3540–3545, Hans Jarett J. Ong et al. "Redefining the Shortest Path Problem Formulation of the Linear Non-Gaussian Acyclic Model: Pairwise Likelihood Ratios, Prior Knowledge, and Path Enumeration". In: *arXiv preprint arXiv:2404.11922* (2025). URL: https://arxiv.org/abs/2404.11922.

[31] 陈琦琦. "基于 Copula 熵和偏秩相关系数的时序因果网络构建算法及其应用". 硕士学位论文. 合肥工业大学, 2023, Jing Yang and Xinzhi Rao. "Copula Entropy Based Causal Network Discovery from Non-stationary Time Series". In: *Pattern Recognition*. Springer Cham, 2025, pp. 115–131.

Growth of Theory

Machine Learning

- structure learning[32], naive Bayes[33], clustering[34], nonlinear PCA[35], decision trees[36], reinforcement learning[37].

[32] Luiz Desuó Neto et al. "Learning Continuous Decomposable Models Using Mutual Information and Statistical Copulas". In: *Entropy* 28.3 (2026), p. 293. ISSN: 1099-4300. DOI: 10.3390/e28030293. URL: https://www.mdpi.com/1099-4300/28/3/293, Luiz Desuó Neto. "Untying the Gordian knot of Bayesian inference in Markov networks". PhD thesis. Universidade de São Paulo, 2025.

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[34] 刘磊 et al. "基于非线性相关性和复杂网络的径流相似性分区". In: 水科学进展 33.3 (2022), pp. 442–451, Francesca Condino. "La divergenza di Jensen-Shannon nell'algoritmo di clustering dinamico per dati descritti da distribuzioni multivariate". PhD thesis. Università degli Studi di Napoli Federico II, 2009.

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Growth of Theory

Neural Networks and Deep Learning

- graph neural networks[38], IEGAIN[39], neural collapse[40], neural network pruning[41], and tree-copula variational autoencoders[42].

[38] 陈佳雷 et al. "一种基于多要素注意力时空图卷积网络的径流预报方法". Pat. CN117151285A. CN117151285A. 2023, 汤宇飞. "基于脉搏波的糖尿病和高血压诊断算法研究". 硕士学位论文. 中国矿业大学, 2023.

[39] 武昊. "基于深度学习的化工过程软测量建模方法研究". 博士学位论文. 北京化工大学, 2023.

[40] Siwei Wang and Stephanie E Palmer. "Towards understanding neural collapse in supervised contrastive learning with the information bottleneck method". In: *arXiv preprint arXiv:2305.11957* (2024). URL: https://arxiv.org/abs/2305.11957.

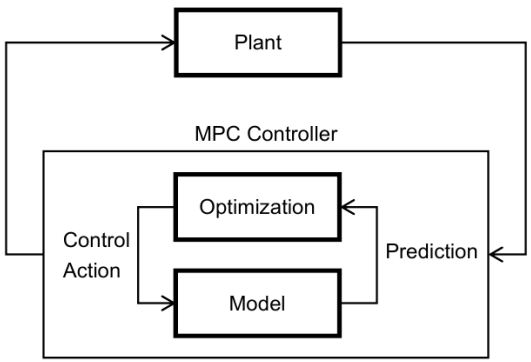
[41] Zihao Jiang et al. "CoPruning: Exploring the Parameter-Gradient Nonlinear Correlation for Neural Network Pruning Using Copula Function". In: *OpenReview.net* (2024). URL: https://openreview.net/forum?id=Yqqa9aNwB0.

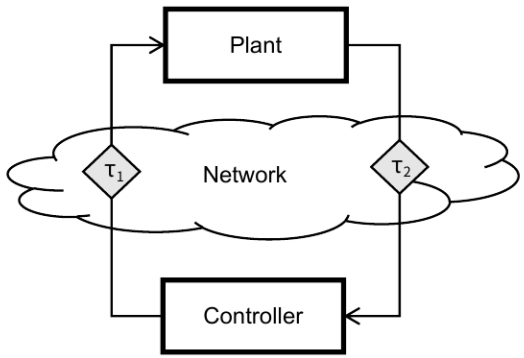
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Growth of Theory

Control Theory

- model prediction controller[43] and sliding mode controller[44].





[43] Zhiwei Li et al. "Passenger-centric model predictive control for indoor temperature to facilitate energy-efficient airport terminal". In: *Building and Environment* 283 (2025), p. 113344. ISSN: 0360-1323. DOI: https://doi.org/10.1016/j.buildenv.2025.113344. URL: https://www.sciencedirect.com/science/article/pii/S0360132325008200.

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Growth of Theory

Optimization

- CE based evolutionary algorithm[45], CE based estimation of vine copula distribution algorithm[46], and CE based grey wolf optimization[47].

[45] Stuart William Card. "Towards an Information Theoretic Framework for Evolutionary Learning". PhD thesis. Syracuse University, 2011.

[46] 王筱萍. "基于分导 Copula 函数的分布估计算法研究". 博士学位论文. 兰州理工大学, 2013.

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Growth of Theory

Approximation Theory

- B-spline approximation[48].

[48] 張🄵棠. "基於 Copula Entropy 的變數選取方法與節點選取方法". 硕士学位论文. 国立政治大学, 2024.

Growth of Theory

Signal Processing

- copula component analysis[49].

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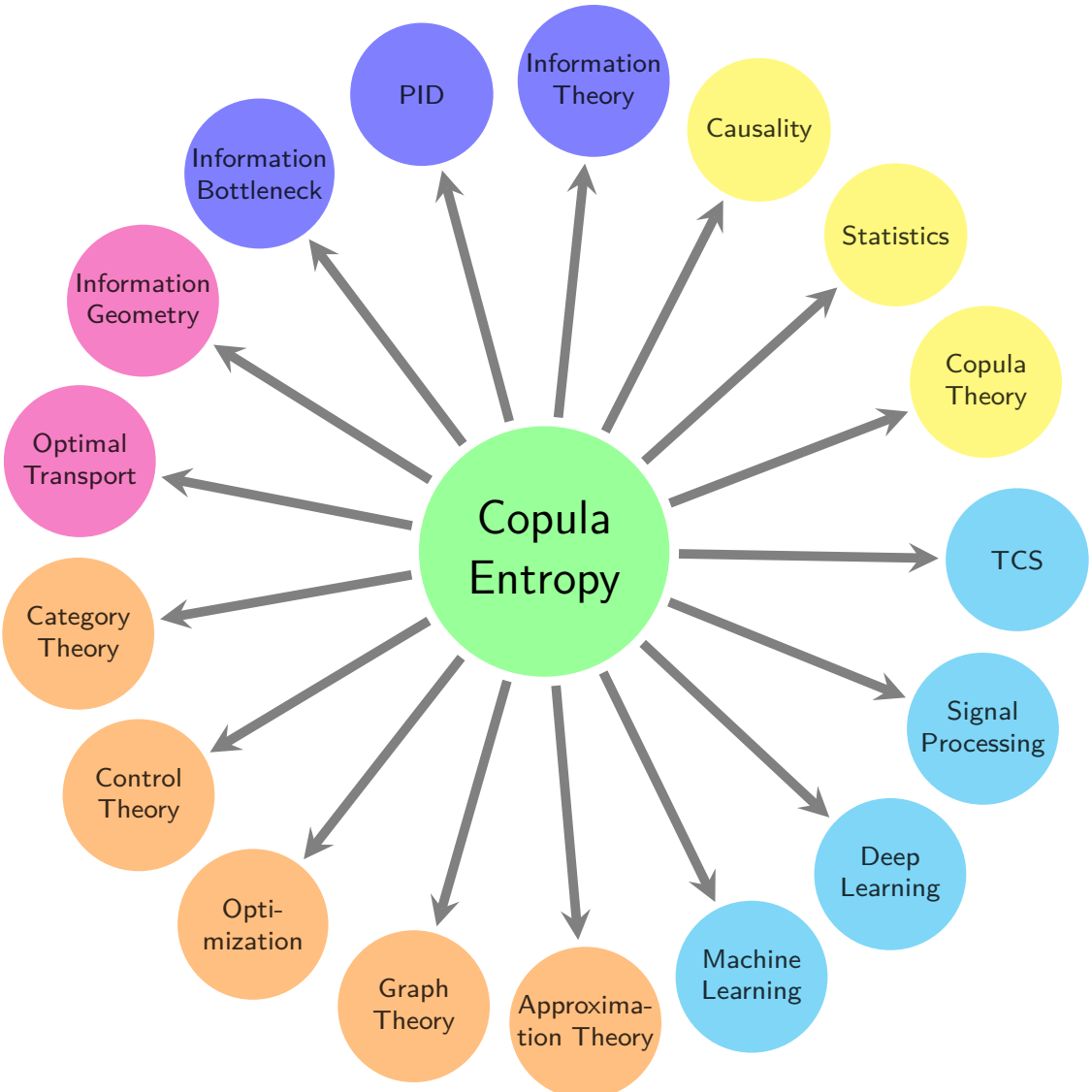
Growth of Theory

Theoretical Computer Science

- NP problem[50].

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Growth of Theory



Monograph

Jian Ma. "Copula Entropy: Theory and Applications". In: *arXiv preprint arXiv:2025.18168* (2025)

Thanks!

Q&A